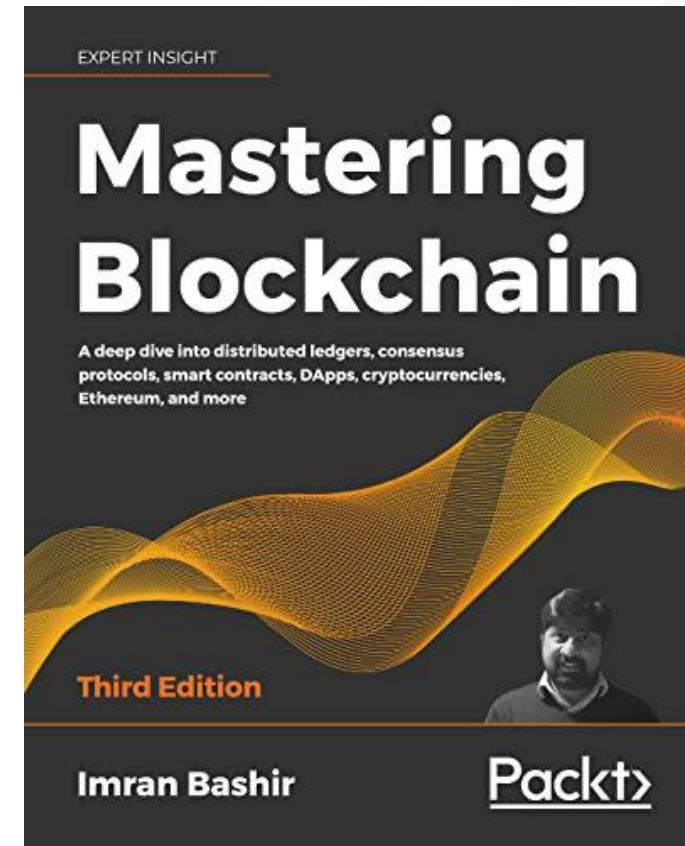


Mastering Blockchain

Third Edition

Chapter 11, Ethereum 101



Outline

- The Ethereum network
- Components of the Ethereum ecosystem
- The Ethereum Virtual Machine (EVM)
- Native contracts



Ethereum – an overview



ETHEREUM: A SECURE DECENTRALISED GENERALISED TRANSACTION LEDGER PETERSBURG VERSION 4ea7b96 – 2020-06-08

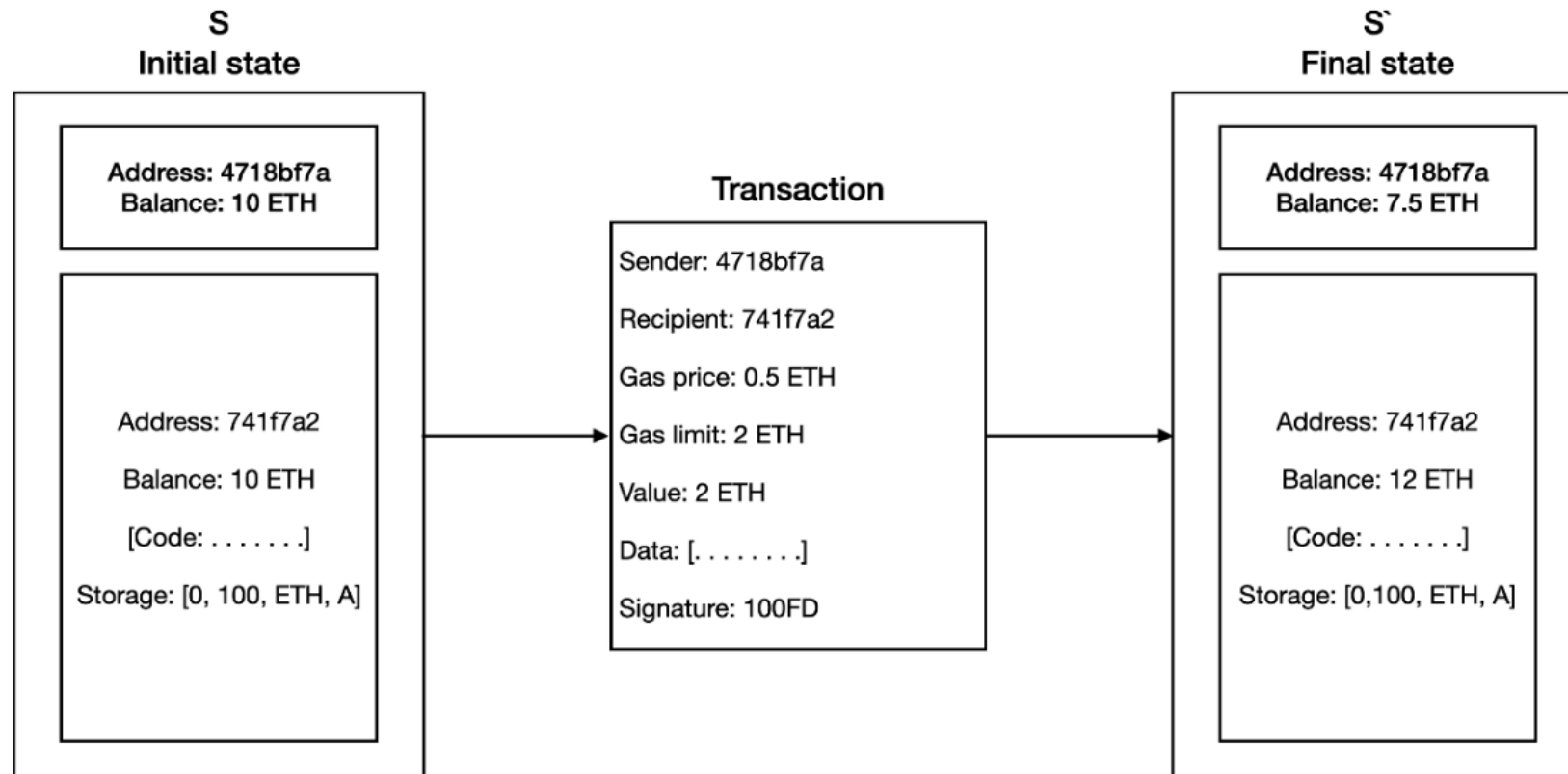
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ABSTRACT. The blockchain paradigm when coupled with cryptographically-secured transactions has demonstrated its utility through a number of projects, with Bitcoin being one of the most notable ones. Each such project can be seen as a simple application on a decentralised, but singleton, compute resource. We can call this paradigm a transactional singleton machine with shared-state.

Ethereum implements this paradigm in a generalised manner. Furthermore it provides a plurality of such resources, each with a distinct state and operating code but able to interact through a message-passing framework with others. We discuss its design, implementation issues, the opportunities it provides and the future hurdles we envisage.

The Ethereum blockchain

On the Ethereum blockchain, a genesis state is transformed into a final state by executing transactions incrementally. The final transformation is then accepted as the absolute undisputed version of the state. Below, the Ethereum state transition function is shown, where a transaction execution has resulted in a state transition.



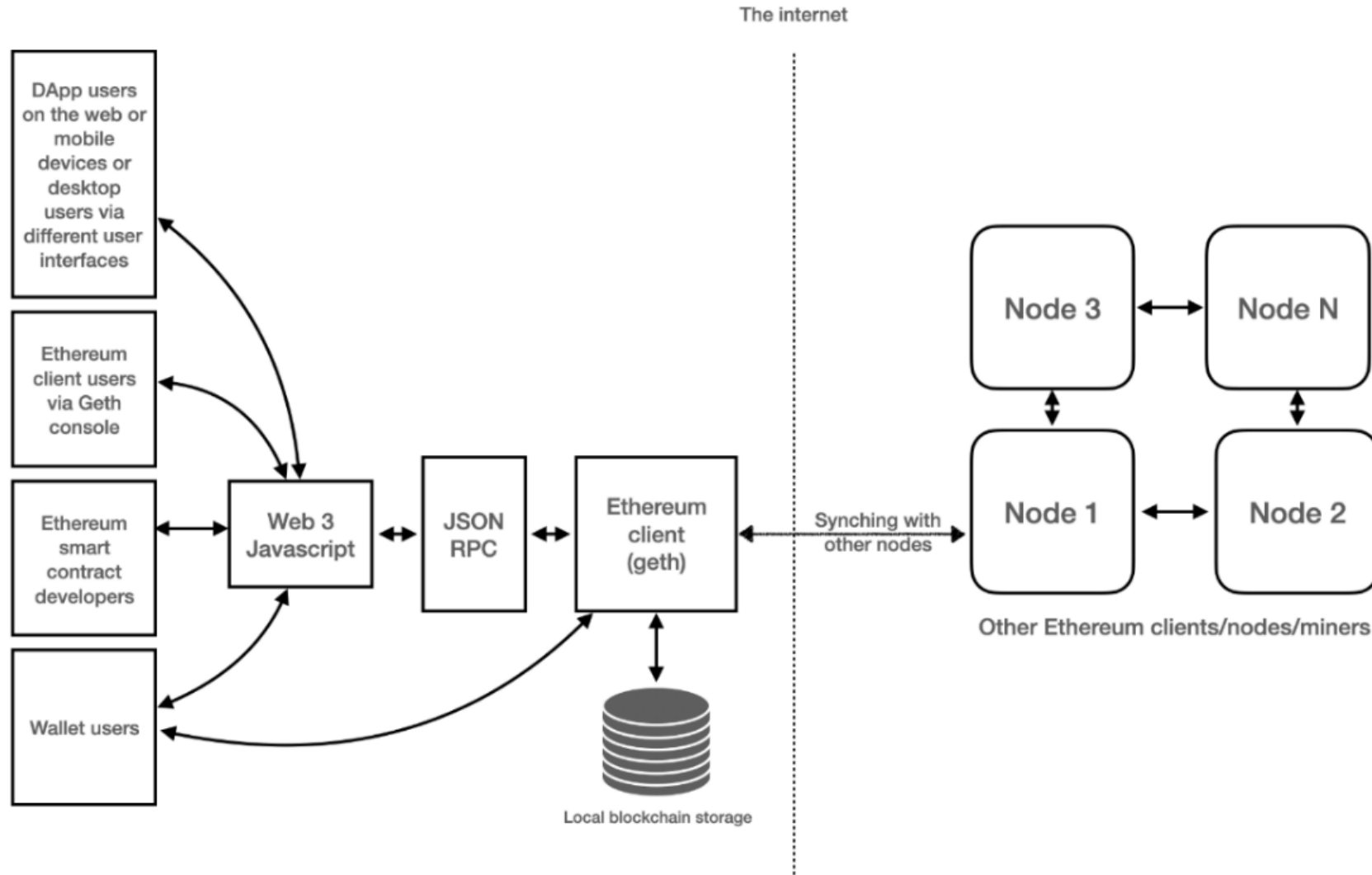
Ethereum – a user's perspective



- The most common use case for the Ethereum chain is to send or receive payments
- To do so, the user signs the transaction and submits it, which propagates on the network, at which point miners pick it up, verify it, and begin Proof of Work (PoW). If PoW is successful, the block with the transaction is finalized and propagated, and a new block is added to the chain
- To send and receive transactions, wallet software is used: for example, mobile wallets are used on mobile devices

Ethereum's high level architecture

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Elements of the Ethereum ecosystem



- Keys and addresses
- Accounts
- Transactions and messages
- Ether cryptocurrency/tokens
- The EVM
- Smart contracts and native contracts

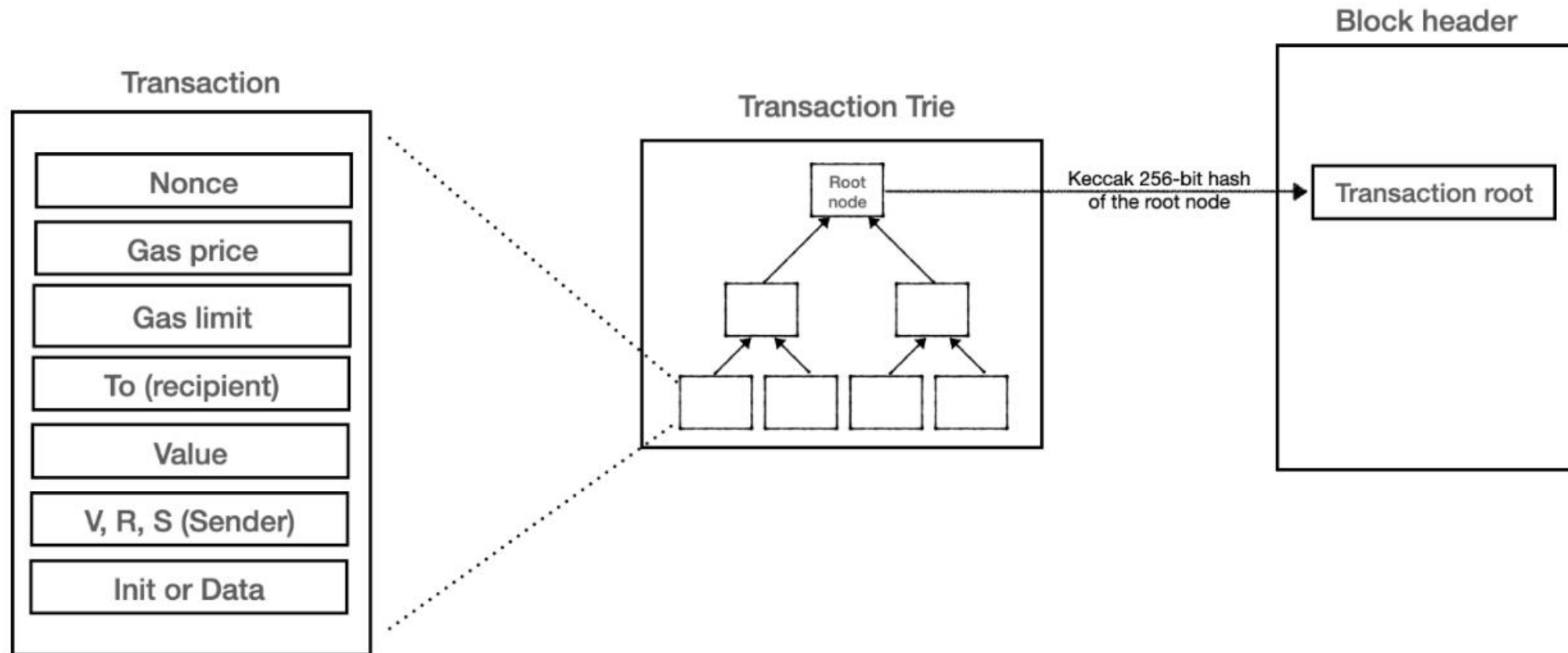
Types of accounts



- EOAs: Externally Owned Accounts. **User accounts** represented by an address
- CAs: **Contract accounts**. Created as a result of deployment of a smart contract—also represented by an address

Transaction and transaction trie

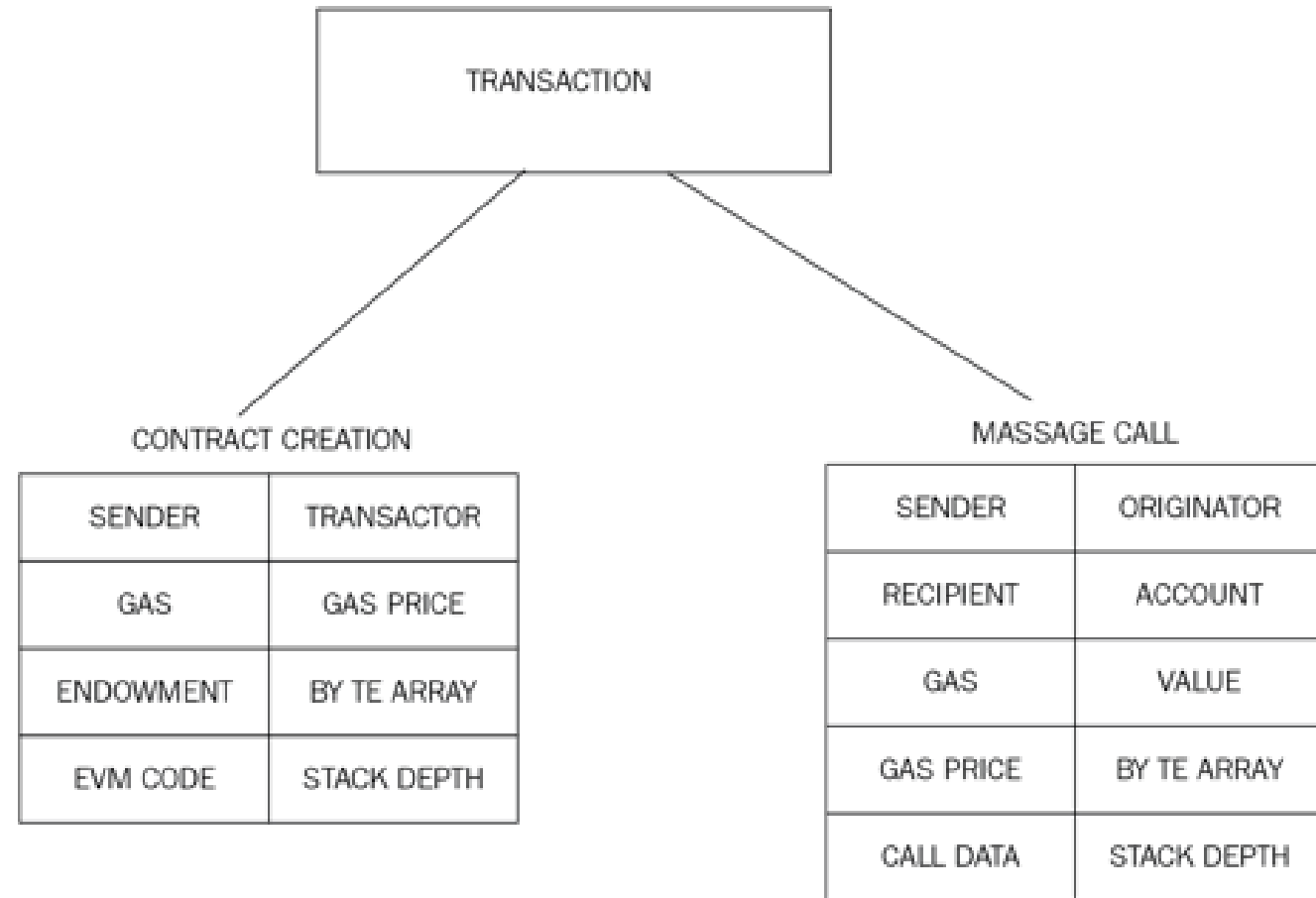
A transaction in Ethereum consists of a number of fields, as shown here, along with transaction trie. The diagram also shows the relationship between the transaction trie and the block header.



Types of transactions

There are three types of transactions:

- **Contract creation**
- **Call**
- **Value transfer**

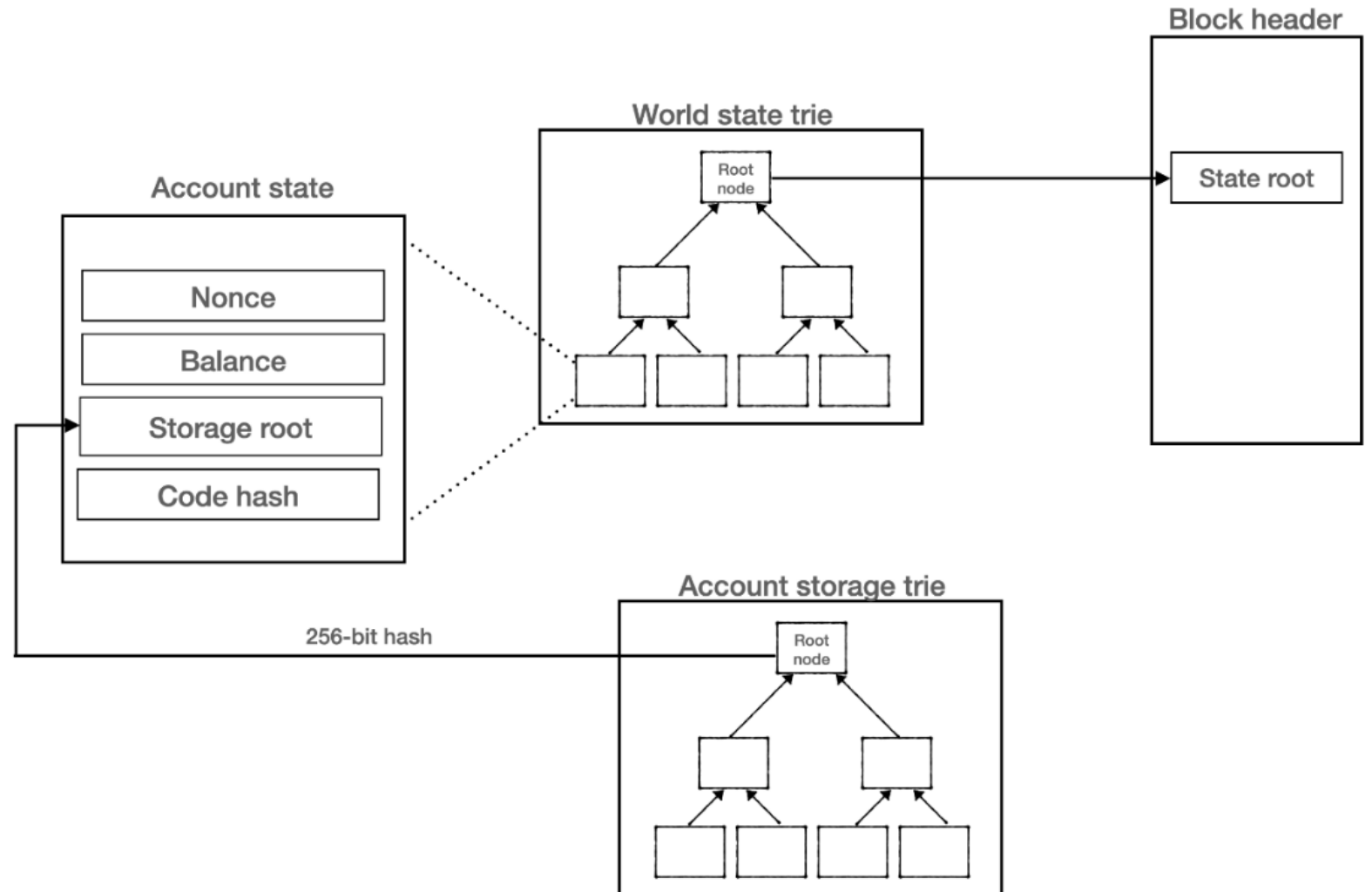


The diagram shows *contract creation* and *message call* transactions, with required fields

Account state and storage trie

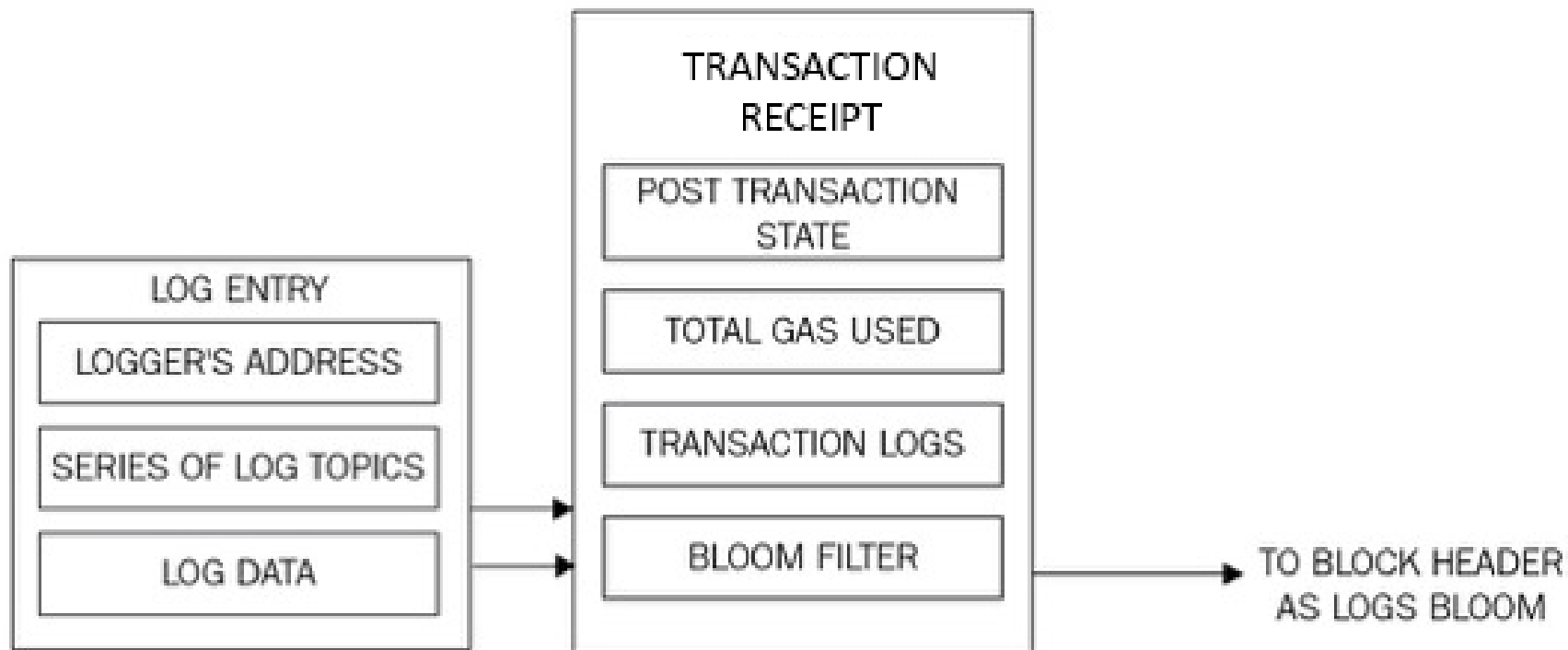
The diagram shows the fields contained within the account state, and how the various elements are contained within the world state trie:

- World state trie
 - State root
 - Account state
 - Account storage trie



Transaction receipts

- Transaction receipts are generated as a result of a transaction execution
- Logs are also updated accordingly
- Both data structures contain several fields, as shown below:

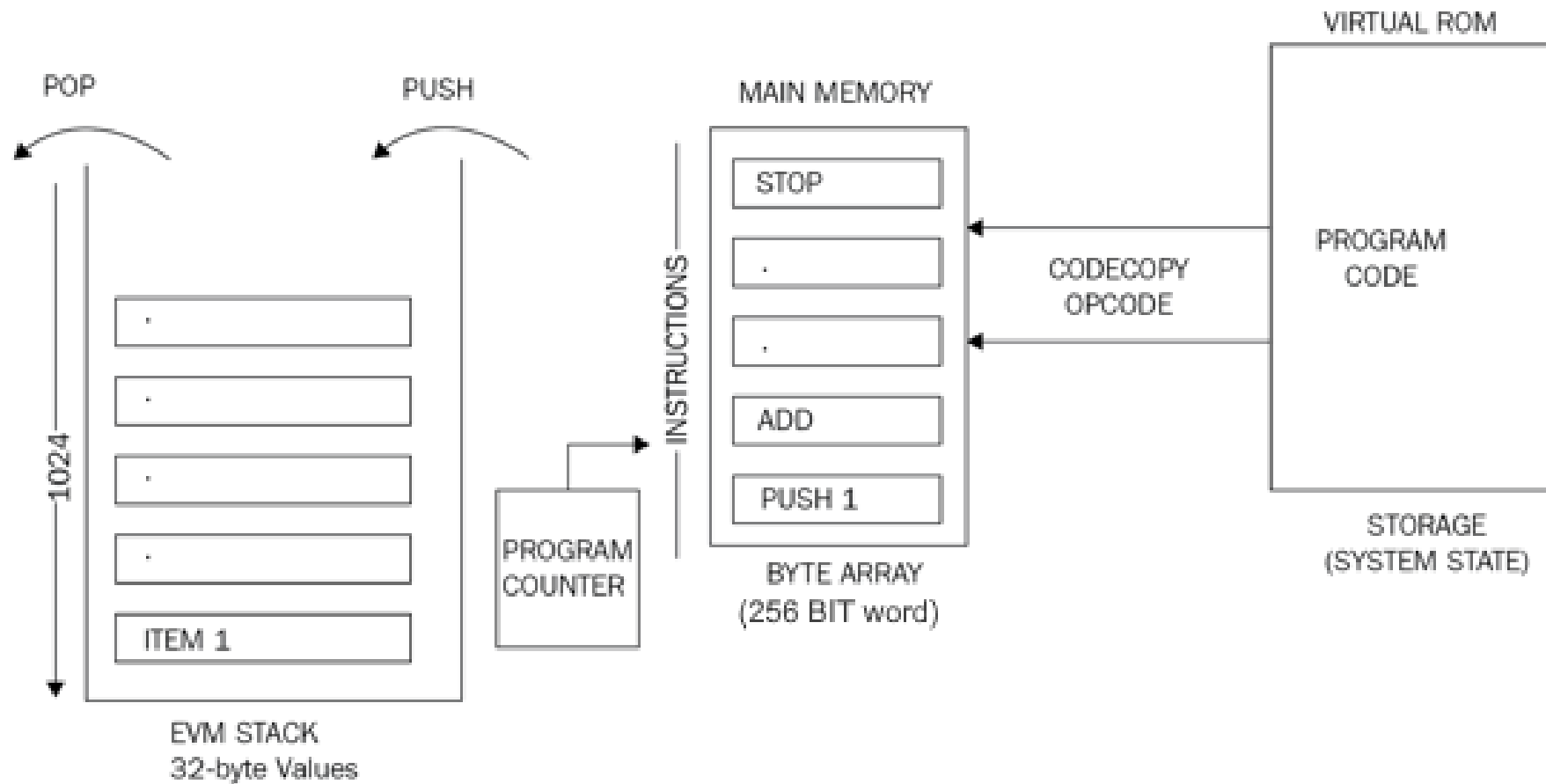


The Ethereum Virtual Environment (EVM)



- Stack size based on LIFO queue: Last In, First Out.
- 1024 stack depth limit
- **Turing complete** but limited by gas, making it quasi-Turing complete
- Big-endian design
- Storage available to EVM
 - Memory
 - Storage
 - Stack

EVM operation design



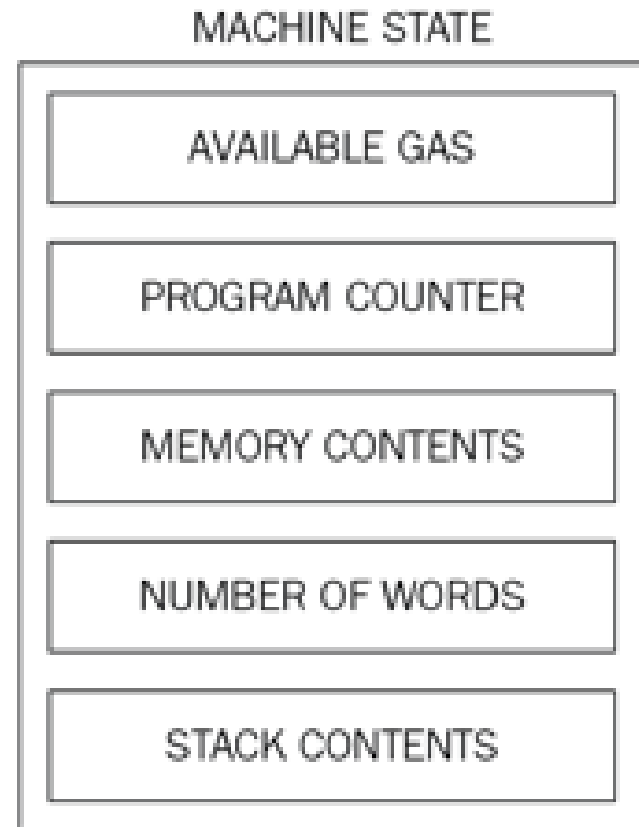
Execution environment

Ethereum's execution environment consists of several elements, as shown here:

Address of code owner
Sender address
Gas price
Input data
Initiator address
Value
Bytecode
Block header
Message call depth
Permission

Machine state

Machine state is a tuple comprising several fields, as shown here:



Native contracts

There are nine **precompiled contracts** or **native contracts** in the Ethereum Istanbul release:

- The elliptic curve public key recovery function
- The SHA-256-bit hash function
- The RIPEMD-160-bit hash function
- The identity/datacopy function
- Big mod exponentiation function
- Elliptic curve point addition function
- Elliptic curve scalar multiplication
- Elliptic curve pairing
- Blake2 compression function 'F'

Exercise

- Read the Ethereum yellow paper:

<https://ethereum.github.io/yellowpaper/paper.pdf>

Summary

In this presentation, we explored:

- The Ethereum network
- Components of the Ethereum ecosystem
- The Ethereum Virtual Machine (EVM)
- Native contracts